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from pathlib import Path
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

data_path = Path("../data/insurance.csv")

if not data_path.exists():
    raise FileNotFoundError(f"File Not Found: {data_path}")

df = pd.read_csv(data_path)
print(f"Shape: {df.shape}")
print("Data Loaded Successfully!")

summary = pd.DataFrame({
    'dtypes': df.dtypes.astype('str'),
    'non_null': df.count(),
    'null_count': df.isnull().sum(),
    'null_pct': (df.isnull().sum() / len(df) * 100).round(2),
    'unique_count': df.nunique(),
    'unique_pct': (df.nunique() / len(df) * 100).round(2)
})

print("------Summary------")
print(summary)

num = df.select_dtypes(include='number')
num_summary = num.describe().T
num_summary['skew'] = num.skew()
num_summary['kurtosis'] = num.kurtosis()
num_summary['range'] = num.max() - num.min()
num_summary['CV'] = (num.std() / num.mean()).round(3)

print("-----Numerical Summary-----")
print(num_summary)

print("-----Categorical Value Counts-----")

cat_cols = df.select_dtypes(
    include=['object', 'string', 'category']
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    include=['object', 'string', 'category']
).columns

for col in cat_cols:
    print(f"----{col}------")
    print(df[col].value_counts())
    print(df[col].value_counts(normalize=True).round(4) * 100)

constant_cols = df.columns[
    df.nunique(dropna=False) == 1
].tolist()

print(f'Constant Columns: {constant_cols if constant_cols else "none✅"}')

id_like = df.columns[
    df.nunique() == len(df)
].tolist()

print(f"ID Like Columns: {id_like if id_like else 'none✅'}")

high_null = df.columns[
    df.isnull().mean() > 0.3
].tolist()

print(f"High Null Columns: {high_null if high_null else 'none✅'}")

print("-----Domain Sanity Checks-----")

domain_sanity = {
    'Age in [0, 120]': df["age"].between(0, 120).all(),
    'BMI in [10, 70]': df["bmi"].between(10, 70).all(),
    'Charges >= 0': (df["charges"] >= 0).all(),
    'Children >= 0': (df["children"] >= 0).all(),
    'Sex Values Valid': set(df["sex"].dropna().unique()) <= {'male', 'female'},
    'Smoker Values Valid': set(df["smoker"].dropna().unique()) <= {'yes', 'no'}
}

for check, result in domain_sanity.items():
    print(f"{check:<22}: {result}")
```